



LLRPReaderUI_WPF UI User Manual

1. Overview

1.1 Software Introduction

LLRPReaderUI_WPF is a professional RFID reader control and testing tool based on the LLRP protocol. It communicates with readers through LLRPSdk and supports device configuration, inventory, memory read/write, and advanced security operations under standard LLRP.

1.2 Runtime Environment

- OS: Windows 10 / Windows 11
- Runtime: .NET 9.0 Runtime
- Network: Reader must be reachable on the same subnet or via routing

1.3 Feature Summary

Module	Description
Device Connection	IP:Port connection, auto-parses device capabilities (model, firmware, antennas, RF params)
Settings	Keepalive, Session, Tag Population Estimate, RF Mode, hop tables, state-aware inventory, hold reports on disconnect, per-antenna config, factory reset
GPIO	GPI enable & live state, GPO target level set/apply
Inventory Config	Auto start/stop, dual filters, tag select filters, report mode (incl. BatchAfterStop) & fields, attached data
Inventory	Real-time inventory, duration, stats (reports/tags/unique), column toggle, manual buffered pull
Read/Write	EPC/TID target matching, 4 memory banks, optional BlockWrite, 5s timeout

Module	Description
Advanced Ops	Lock/Unlock/Permalock/Permaunlock, Kill, BlockErase, 5s timeout
Logs	Operation/LLRP/Raw logs, batched UI refresh
LLRP Messages	SQLite history, multi-filter query, message tree parse & export

2. Quick Start

2.1 Connect Device

1. In **Device Connection**, enter reader IP or IP:Port (supports 3 recent endpoints).
2. Click **Connect**. On success, the status bar shows connection state and MAC. Capabilities load below.
3. If the reader is inventorying, the app stops it before initialization.

2.2 One-Click Inventory

1. Navigate to **Inventory**.
2. Click **Start Inventory**.
3. Tags appear in real time; totals show report count, total tags, and unique tags.

3. Module Details

3.1 Device Connection & Capabilities

After connection, the full FeatureSet is shown:

- Core info: ModelNumber, ReaderModel, manufacturer ID (with name), firmware, antenna/GPI/GPO counts.
- Protocol: CommunicationsStandard, country code (with name), max sequences/ops, tag access and filtering capabilities.
- Memory ops: IsMultiwordBlockWriteAvailable, IsMultiwordBlockEraseAvailable.

- RF: IsHoppingRegion, power levels, sensitivity levels, frequency points, RF modes.

Keepalive timeout protection: if the reader stops responding to keepalive, the app disconnects and shows timeout.

3.2 Settings

Core actions: **Get, Save, Factory Reset.**

- Keepalive: enable/disable and interval (ms).
- RF mode & frequency:
 - RF Mode uses Pn prefix and mode id (e.g., P0-0, P1-2) with detailed params.
 - Hop tables: select HopTable ID; in hopping regions (FCC/ETSI) the reader cycles table frequencies; a single channel index is usually ignored.
 - Channel index: set Channel Index and show current frequency (fixed frequency in non-hopping region).
- Inventory strategy:
 - Session: 0/1/2/3.
 - TagPopulationEstimate affects inventory efficiency.
 - State-Aware: if supported, configure Target (A/B) and Search Mode (SL/Not_SL).
- Disconnect strategy: Hold Reports On Disconnect.
- Antennas: per-port enable, Tx Power (dBm), Rx Sensitivity (dBm).
- Events: displays ReaderEventNotificationSpec enablement.

3.3 GPIO

- GPI: per-port enable and current level (High/Low/Unknown), save to apply.
- GPO: per-port target level; apply to set and refresh.

3.4 Inventory Config

Core actions: **Read, Save.**

- Auto start:
 - None, Immediate, GPI Trigger, Periodic; periodic supports first delay / UTC time / period.
- Auto stop:

- None, Duration, GPI Trigger, GPI Timeout (Timeout=0 means unlimited).
- Filters:
 - Modes: None, Single, Dual.
 - Standard filters: Filter1/Filter2 by memory bank (EPC/TID/User/Reserved), bit offset/count, mask, Match/NonMatch.
 - Tag Select Filters: dynamic list of Select rules with memory bank, bit offset/count, mask, match/non-match action.
- Report:
 - Modes: Individual, WaitForQuery, BatchAfterStop.
 - Fields: PC, CRC, first/last seen time, antenna port, channel, Peak RSSI, seen count.
- Attached data: read a memory bank (default TID) with word pointer/count (default 6) and access password.

3.5 Inventory

- Actions: Start, Stop, Clear, Manual Pull.
- Stats: total reports, total tags (with duplicates), unique tags, and duration.
- Manual Pull: only available when report mode is WaitForQuery.
- Column visibility mirrors report field options.
- Tag fields: receive time, EPC, antenna port, channel frequency (MHz), Peak RSSI (dBm), seen count, PC, CRC, first/last seen time (UTC), attached data.
- Capacity: list shows up to 500 rows (oldest removed).

3.6 Read/Write

- Target: match by EPC (bit 32) or TID (bit 0).
- Memory banks: User, TID, Reserved, EPC.
- Read: specify word pointer/count.
- Write: hex data length must be a multiple of 4 (word aligned).
- BlockWrite: available if the reader supports it.
- Access password: 32-bit hex (default 00000000).
- 5s timeout protection.
- Attached data restore: inventory/attached state is saved before ops and restored after.

3.7 Advanced Tag Ops

- Target: same as read/write (EPC/TID).
- Functions:
 - Lock: Lock/Unlock/Permalock/Permaunlock for KillPassword, AccessPassword, EPC, TID, User.
 - Kill: 32-bit kill password, 8 hex chars.
 - Block Erase: only if supported.
- Automation: 5s timeout; attached data state saved/restored; clears op sequences.

3.8 Logs

- Types: operation, LLRP message, raw frame.
- Toggle display per type.
- Batched UI refresh every 200ms.
- Capacity: 500 rows per log type.

3.9 LLRP Message Browser

Loads history from SQLite (`llrp_rawframes.db`).

- Filters: time range, direction (All/RX/TX), message type, device ID, text search, clear filters.
- Parsing: select a record to show message tree and raw hex.
- Actions: Refresh (last 5000), Export (tree to `.txt` , saved to app working directory), Clear (delete all records).

3.10 Status Bar

Shows device state, inventory state, antenna state, GPI/GPO state, and MAC. **Refresh Status** triggers manual query.

3.11 Theme & Language

- Chinese/English UI switching.
- Light/Dark theme switching.
- Navigation and prompts update immediately after switching.

4. FAQ

Q: Why is Channel Index ineffective in hopping mode?

A: In hopping regions the reader cycles hop tables; single channel index is usually ignored.

Q: Advanced or Read/Write shows Timeout ?

A: Operations have a 5s timeout. Ensure tag is close and filters are not too strict.

Q: Difference between Receive Time and FirstSeenTime?

A: Receive time is the PC local time when the report arrived; FirstSeenTime is the reader UTC timestamp.

Q: Manual Pull button disabled?

A: Only available when report mode is WaitForQuery.

Q: BlockWrite / BlockErase disabled?

A: Depends on reader capability; unsupported devices disable those buttons automatically.

Appendix

- Local storage:
 - LLRP raw frame DB: %LocalAppData%\LLRPReaderUI_WPF\llrp_rawframes.db
 - Recent endpoints: %LocalAppData%\LLRPReaderUI_WPF\recent-endpoints.json
 - Log files: written by Serilog
 - Raw frame retention: older than 7 days is purged on startup
- Version:
 - App version: 1.0.15-Alpha-260513
 - Document version: 1.4
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